

CONTACT

+61 423 681 009

caitlin@auspatious.com

Adelaide, Australia

auspatious.com

Research Contributions

SKILLS

- Chair, Documentation Subcommittee | Open Data Cube Steering Council | 2024 – Present (member since 2023)
- Steering Council Member | Open Data Cube | 2019 – Present (Chair, 2022–2023)

EDUCATION

- Doctor of Philosophy in Astrophysics | Swinburne University of Technology
- Bachelor of Science in Physics (First Class Honours) | University of Queensland

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CAITLIN ADAMS

SENIOR DATA SCIENTIST

Caitlin has spent her career turning satellite data into insights about our planet. Her background in physics makes her an excellent problem solver, with deep expertise in turning data into meaningful information. Following a PhD in astrophysics, Caitlin has spent eight years building machine learning and remote sensing pipelines across Australia, Antarctica, and Africa. Caitlin is a member of the Open Data Cube Steering Council, where she volunteers her time supporting the global Earth observation community.

FEATURED PROJECTS

SAR DATA PROCESSING AND PRODUCT DEVELOPMENT FOR AUSTRALIA AND ANTARCTICA

GEOSCIENCE AUSTRALIA

- Led development of processing pipelines converting Level-1 Sentinel-1 SAR data into Level-2 normalised radar backscatter, using open-source tools including ISCE3 and PyroSAR.
- Designed an AWS pipeline in Terraform supporting both ongoing processing and full historical reprocessing of the Sentinel-1 archive.
- Co-developed a Python library for handling digital elevation models, including complex cases at the antimeridian.
- Worked with university and government stakeholders to build prototype products, including automated mapping of land-fast and blue ice, and supported certification of the product's STAC metadata against the CEOS-ARD specification.

DIGITAL EARTH AFRICA WATERBODIES

FrontierSI

- Led adaptation of Digital Earth Australia's Waterbodies pipeline for continental-scale application across Africa, improving computational efficiency and reducing AWS processing costs.
- Designed an Argo Workflows architecture to process 30+ years of Landsat-derived Water Observations from Space data into time series, with automated weekly updates.
- Worked with Digital Earth Africa staff to understand user needs and co-design a computationally efficient mapping approach, now a core program service.

TREE CANOPY MAPPING

FrontierSI

- Led technical development of a suite of tree-cover products, applying deep learning (convolutional neural networks) to 10–20cm aerial RGB imagery across Victoria.
- Used AWS Batch and Docker to process 37 terabytes of aerial imagery stored in AWS S3, building a scalable, cloud-native workflow.



LANGUAGES

- Python
- C++

PROFESSIONAL AFFLIATIONS

- **Chair, Documentation Subcommittee** | Open Data Cube Steering Council | 2024 – Present (member since 2023)
- **Steering Council Member** | Open Data Cube | 2019 – Present (Chair, 2022–2023)

AWARDS

- Early-Career Contribution to the Australian Earth Observation Community Award | Earth Observation Australia | 2022
- Leading Women in Machine Learning for Earth Observation | Radiant Earth | 2022

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CAITLIN ADAMS

SENIOR DATA SCIENTIST

SENIOR DATA SCIENTIST | FRONTIERSI

- Led development of continental-scale remote sensing products from Landsat and Sentinel-2 optical imagery, including adapting Digital Earth Australia's Waterbodies pipeline for Digital Earth Africa.
- Improved computational efficiency of processing workflows using Argo, reducing processing time and AWS cloud costs.
- Provided Agile project management and stakeholder communication across large-scale product development projects.
- Consulted with other organisations on developing Earth observation products, and supported the professional development of junior Data Scientists.

DATA SCIENTIST | FRONTIERSI

- Led technical development of a suite of tree-mapping products for the Victorian Department of Environment, Land, Water and Planning, applying machine learning to aerial imagery on AWS.
- Applied machine learning to Earth observation data to develop informative, decision-ready datasets.
- Developed educational resources with Digital Earth Australia and Digital Earth Africa to help end users access and apply Earth observation data.

DATA ANALYST AND MODELLER | MEZO RESEARCH

- Developed analytic tools and statistical/mathematical models for natural and human resource management projects, including clients such as Victoria Police and the Victorian Fisheries Authority.
- Communicated modelling approaches and findings directly to clients to inform management decisions.
- Oversaw the company's Agile development process through Git and Jira.